

Sepsis Biomarkers Analysis Using SERS-Based Sensor

A Project Report submitted by
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in partial fulfillment of the requirements for the award of the degree of
Masters in Medical Technologies



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Declaration

I hereby declare that the work presented in this Project Report titled “**Sepsis Biomarkers Analysis Using SERS-Based Sensor**”, submitted to the Indian Institute of Technology Jodhpur and All India Institute of Medical Sciences Jodhpur in partial fulfilment of the requirements for the award of the degree of **Masters in Medical Technologies**, is a bonafide record of the research work carried out under the supervision of **Prof. Ajay Agarwal** and **Dr. Neeraj Gupta**.

The contents of this Project Report in full or in parts have not been submitted to, and will not be submitted by me to, any other Institute or University in India or abroad for the award of any degree or diploma.



Signature

Sukesh P

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Certificate

This is to certify that the Project Report titled "**Sepsis Biomarkers Analysis Using SERS-Based Sensor**", submitted by **Sukesh P (M24IM1007)** to the Indian Institute of Technology Jodhpur and All India Institute of Medical Sciences Jodhpur, is a bonafide record of the research work carried out under my supervision. To the best of my knowledge, the report has not been submitted to any other Institute or University for the award of any degree or diploma.


Signature

Prof. Ajay Agarwal


Signature

Dr. Neeraj Gupta

Abstract

Sepsis is a severe and life-threatening condition arising from an uncontrolled systemic immune response to infection. It continues to be one of the major causes of mortality worldwide, particularly in critical care settings and neonatal intensive care units (NICUs). Current diagnostic approaches rely heavily on host-response biomarkers such as C-reactive protein (CRP), procalcitonin (PCT), and interleukins, which lack bacterial specificity and often lead to empirical antibiotic treatment without pathogen confirmation.

In this work, a **Surface-Enhanced Raman Spectroscopy (SERS)-based nano-diagnostic platform** is developed for direct detection of **bacterial structural biomarkers**. Instead of detecting host inflammatory molecules, this project focuses on identifying bacterial-specific molecules present on the surface of pathogens. The selected biomarkers include:

- **Peptidoglycan (PGN)** – universal bacterial cell wall component
- **Lipopolysaccharide (LPS)** – Gram-negative bacterial marker
- **Lipoteichoic Acid (LTA)** – Gram-positive bacterial marker

A **bimetallic Ag/Au nanostructured SERS substrate** fabricated on silicon is used to generate localized surface plasmon resonance (LSPR) hotspots, significantly enhancing Raman scattering signals. Using a 785 nm excitation wavelength, characteristic Raman fingerprints of the three biomarkers were analyzed under optimized experimental conditions.

This approach offers significant advantages including rapid detection (<30 minutes), label-free analysis, bacterial specificity, and Gram classification capability. The platform is being developed as a potential point-of-care diagnostic tool that can support early clinical decision-making in sepsis, reduce inappropriate antibiotic usage, and improve patient outcomes.

1. COMPREHENSIVE OVERVIEW OF THE STATE OF THE PROBLEM & EXISTING SOLUTIONS

Sepsis is a life-threatening medical condition caused by a dysregulated immune response to infection, leading to organ dysfunction and, if untreated, multi-organ failure and death. According to global health studies, sepsis affects millions of people annually and remains one of the leading causes of mortality in intensive care units (ICUs) and neonatal intensive care units (NICUs), particularly in low- and middle-income countries such as India.

The primary clinical challenge in sepsis management is **early and accurate diagnosis**, as the first few hours after symptom onset are crucial for patient survival. Studies have shown that for every hour delay in appropriate antibiotic therapy, mortality increases significantly. However, current diagnostic approaches suffer from fundamental limitations that delay decision-making processes.

1.1 Limitations of Blood Culture

Blood culture is considered the gold standard for bacterial identification. However, it suffers from multiple drawbacks:

- Turnaround time ranges from 24 to 72 hours.
- Sensitivity is reduced in cases of low bacterial load.
- Prior antibiotic administration can cause false-negative results.
- Requires sophisticated laboratory setup and trained personnel.

Due to these limitations, clinicians often start empirical broad-spectrum antibiotic therapy, which contributes to antimicrobial resistance (AMR) and unnecessary toxicity in patients.

1.2 Limitations of Host Response Biomarkers

Currently used biomarkers such as CRP, procalcitonin (PCT), IL-6, and IL-8 are indirect indicators of infection. Although they provide faster results, they suffer from:

- Lack of specificity towards bacterial infections.
- Elevated levels in non-infectious inflammatory conditions.
- Inability to identify pathogen type or Gram classification.

This results in diagnostic uncertainty and often fails to guide targeted antimicrobial therapy.

1.3 Limitations of Molecular Diagnostics

Advanced molecular tools such as PCR and Next-Generation Sequencing (NGS) provide faster pathogen detection. However:

- They are expensive and infrastructure-intensive.
- Not feasible in most government hospitals and rural healthcare centers.
- They cannot distinguish between live and dead bacteria.
- Not suitable for routine point-of-care deployment.

1.4 Role of SERS in Sepsis Diagnostics

Surface-Enhanced Raman Spectroscopy (SERS) offers a promising solution to bridge this diagnostic gap. SERS enhances weak Raman signals using plasmonic nanostructures, enabling detection of biomolecules at very low concentrations.

Unlike traditional biomarker assays, this project focuses on **direct bacterial detection** by identifying:

- Peptidoglycan (PGN)
- Lipopolysaccharide (LPS)
- Lipoteichoic Acid (LTA)

These are structural biomarkers that exist only in bacteria, enabling high specificity and allowing Gram-positive and Gram-negative classification, which is essential for selecting correct antibiotic therapy.

This makes SERS a promising approach for developing rapid, accurate, and point-of-care diagnostic platforms for sepsis management.

2. INTELLECTUAL PROPERTY (PATENT LANDSCAPING)

A detailed review of existing patents and technological developments was conducted using multiple patent databases including WIPO Patentscope, Google Patents, and USPTO.

While several patents exist in the domains of:

- SERS substrate fabrication
- General biosensors
- Sepsis diagnostic devices

Very few systems approach sepsis from the perspective of **bacterial structural biomarker detection**.

Most patented systems focus on:

- Electrochemical sensing
- Immunoassay-based detection
- PCR-based molecular diagnostics

However, no integrated patented system was found that combines:

- SERS-based sensing
- PGN + LPS + LTA biomarker panel
- Gram differentiation
- Portable point-of-care workflow

This clearly demonstrates that the proposed system fills a significant technological gap. The integration of a bacterial biomarker panel with SERS technology and future machine learning models for automated classification significantly strengthens the novelty of this work.

This also establishes strong commercial and intellectual property potential, especially in sepsis diagnostics where rapid point-of-care solutions are urgently needed.

3. OBJECTIVES & METHODOLOGY

3.1 Detailed Objectives

The objectives of this project extend beyond basic detection and focus on translational potential. These include:

1. Development of reproducible Ag/Au bimetallic SERS substrates suitable for biomedical applications.
2. Characterization of Raman spectral fingerprints of PGN, LPS, and LTA.
3. Development of a standardized sample handling and spectral acquisition workflow.
4. Establishment of spectral differences between Gram-positive and Gram-negative bacteria.
5. Preparation for future machine learning-based automated classification of bacterial infections.
6. Exploration of scalability towards portable point-of-care diagnostic systems.

These objectives ensure that the research is not limited to theoretical development but also considers clinical applicability.

3.2 Expanded Methodology

The methodology involved both experimental and analytical components.

Bacterial biomarkers PGN, LPS, and LTA were selected based on their biological relevance, structural uniqueness, and Raman activity.

Ag/Au bimetallic substrates were fabricated using sequential deposition techniques. Silver nanoparticles were deposited on silicon substrates followed by gold coating to improve stability and plasmonic properties.

Characterization was performed using:

- FESEM for surface morphology
- EDS for elemental composition
- UV-Vis spectroscopy for plasmon resonance behavior

Raman spectra were recorded using a 785 nm excitation laser under optimized parameters.

Spectral preprocessing included noise filtering, baseline subtraction, normalization, and peak extraction using computational tools such as MATLAB/Python.

4. DETAIL OF WORK DONE

The experimental and analytical work was carried out in multiple phases:

1. A detailed literature survey was conducted to identify bacterial biomarkers relevant for sepsis and SERS detection.
 2. The biomarker panel (PGN, LPS, LTA) was finalized based on bacterial specificity and Gram classification relevance.
 3. Ag/Au SERS substrates were fabricated multiple times to achieve optimal reproducibility and signal enhancement.
 4. Raman spectral acquisition was performed using controlled experimental conditions for each biomarker.
 5. Characteristic spectral features were identified and linked with molecular vibrational modes.
 6. A preliminary diagnostic workflow was conceptualized for fast bacterial sepsis detection.
 7. Plans for machine learning integration and clinical validation were prepared.
- This work forms a strong foundation for the next stage of clinical and device-level development.

5. CONCLUSION

This project demonstrates the strong potential of Surface-Enhanced Raman Spectroscopy (SERS) as an advanced diagnostic tool for rapid and direct bacterial sepsis detection using structural bacterial biomarkers. Unlike conventional diagnostic approaches that rely on indirect host-response biomarkers or time-consuming culture techniques, this work adopts a fundamentally different strategy by targeting bacterial cell wall and membrane components themselves. The selection of Peptidoglycan (PGN), Lipopolysaccharide (LPS), and Lipoteichoic Acid (LTA) as the biomarker panel ensures high pathogen specificity and enables Gram-level classification of bacterial infections, which is clinically crucial for initiating appropriate antibiotic therapy.

Through this work, a reproducible Ag/Au bimetallic nanostructured SERS substrate was successfully developed and optimized on a silicon platform. The substrate design enabled localized surface plasmon resonance (LSPR) hotspot formation, significantly enhancing Raman signals of the targeted bacterial biomarkers. The use of 785 nm excitation helped reduce background fluorescence and improved signal-to-noise ratio, making the platform suitable for biological sample analysis. Initial spectral studies confirmed that PGN, LPS, and LTA produce distinguishable and characteristic Raman fingerprints when adsorbed on the SERS substrates. These distinct spectral features provide a reliable basis for bacterial detection as well as differentiation between Gram-positive and Gram-negative infections.

The project also contributes towards addressing the major limitations of current sepsis diagnostics. Traditional blood culture, although considered the gold standard, suffers from long turnaround times and low early sensitivity, while immunoassays based on CRP, PCT, and cytokines lack pathogen specificity. In contrast, the proposed SERS-based platform offers rapid, label-free, and reagent-free detection, reducing dependency on antibodies and expensive consumables. This makes it particularly suitable for point-of-care applications in resource-limited settings where quick decision-making is critical for patient survival.

Beyond technical development, this study establishes a structured workflow for bacterial biomarker detection using SERS, starting from biomarker selection, substrate fabrication, spectral acquisition, and data preprocessing, to biomarker-specific peak analysis. This structured approach allows for easy reproducibility and scalability in future studies. Furthermore, the project has laid the groundwork for future integration of machine learning and pattern recognition algorithms, which will enable automated classification of bacterial infections based on spectral data. Such integration will significantly enhance the clinical usability of the system by reducing dependence on expert interpretation.

From a translational perspective, the outcomes of this project demonstrate feasibility for developing a portable point-of-care diagnostic device. With further refinement, miniaturization of Raman instrumentation, and integration with automated data analysis tools, this technology can be transformed into a compact, low-cost bacterial sepsis diagnostic unit suitable for use in ICUs, emergency departments, and rural healthcare centers. This has strong potential to reduce diagnostic delays, improve patient outcomes, guide rational antibiotic usage, and ultimately contribute towards combating the global challenge of antimicrobial resistance.

In conclusion, this work provides both a scientific and technological foundation for the development of a next-generation bacterial sepsis diagnostic platform. By combining nanotechnology, spectroscopy, and biomedical diagnostics, this project offers a promising pathway towards faster, more specific, and clinically actionable sepsis detection solutions. Future work will focus on validation using real patient samples, improvement of substrate reproducibility across large batches, integration with machine learning algorithms, and development of a functional prototype device for clinical testing and regulatory approval.

6. NOVELTY

The novelty of this project lies in its fundamentally different approach to sepsis diagnosis by focusing on **direct bacterial structural biomarkers** instead of conventional host-response biomarkers such as CRP, PCT, and cytokines. By targeting Peptidoglycan (PGN), Lipopolysaccharide (LPS), and Lipoteichoic Acid (LTA), the system enables direct bacterial detection with Gram-positive and Gram-negative classification, which is clinically critical for selecting appropriate antibiotic therapy.

Unlike existing diagnostic platforms based on immunoassays or molecular techniques, this work demonstrates the application of **Surface-Enhanced Raman Spectroscopy (SERS)** for bacterial biomarker detection. The use of SERS enables label-free and reagent-free detection, eliminating dependency on antibodies and fluorescent tags, thereby reducing cost and system complexity.

A major novelty also lies in the **design and optimization of Ag/Au bimetallic nanostructured SERS substrates** specifically for large bacterial macromolecules such as PGN, LPS, and LTA. The bimetallic architecture provides enhanced plasmonic performance along with improved chemical stability compared to conventional silver-based substrates.

Furthermore, this project enables **Gram-level bacterial classification using Raman spectral fingerprints in a single diagnostic step**, eliminating the need for separate Gram staining and culture procedures. The work also establishes a foundation for future integration of machine learning algorithms for automated spectral classification and decision support in clinical settings.

Finally, the interdisciplinary integration of nanotechnology, spectroscopy, microbiology, and biomedical engineering into a unified diagnostic framework makes this work distinct from existing research and commercial platforms, providing a novel pathway towards rapid, low-cost, and point-of-care sepsis diagnostics.

7. POTENTIAL SOCIETAL & MARKET IMPACT

Sepsis is a major global healthcare challenge due to its high mortality, prolonged hospital stay, and significant economic burden. In India and other low- and middle-income countries, the impact of sepsis is more severe because of delayed diagnosis, limited access to advanced diagnostic infrastructure, and poor availability of rapid point-of-care technologies. The SERS-based bacterial sepsis diagnostic platform developed in this project has the potential to address several of these challenges through faster, more specific, and affordable detection of bacterial infections.

From a clinical perspective, one of the major causes of high sepsis mortality is delayed diagnosis and late initiation of targeted antibiotic therapy. The proposed platform enables rapid detection of bacterial biomarkers such as PGN, LPS, and LTA, drastically reducing diagnostic turnaround time from several days to minutes. This can significantly improve patient outcomes by allowing early therapeutic intervention, reducing ICU mortality, and preventing progression to severe organ failure.

Another important societal contribution of this technology lies in reducing antimicrobial resistance (AMR). The widespread use of broad-spectrum antibiotics without confirmed bacterial diagnosis is a major driver of AMR. By providing direct bacterial detection and Gram classification, this system supports rational, evidence-based antibiotic prescription. This can reduce unnecessary antibiotic usage and contribute towards global efforts to control antimicrobial resistance.

The technology is also highly relevant for rural and resource-limited healthcare settings where access to sophisticated laboratory facilities is limited. With future miniaturization and cost optimization, the system can be deployed as a point-of-care diagnostic device in primary health centers and small hospitals. This will improve access to quality diagnostics in remote regions and help reduce healthcare disparities between urban and rural populations.

From an economic perspective, early and accurate sepsis diagnosis can reduce ICU stay duration and overall treatment costs, thereby lowering the financial burden on both patients and healthcare systems. Shorter hospitalization and targeted treatment can lead to significant cost savings at both individual and national levels.

In terms of market potential, the global sepsis diagnostics market is rapidly expanding due to increased awareness of early detection and rising sepsis incidence. Existing market solutions largely rely on immunoassay or molecular platforms, which are often expensive and infrastructure-dependent. The proposed SERS-based system fills a critical gap by offering a low-cost, rapid, and pathogen-specific alternative, making it suitable for both advanced hospitals and low-resource settings.

In the Indian context, this project aligns well with the vision of developing indigenous medical technologies under initiatives such as "Make in India". It can reduce dependence on imported diagnostic equipment, promote local manufacturing, and support the growth of medical device startups and deep-technology ventures. In the long term, this technology has strong potential for commercialization and large-scale deployment in hospitals, diagnostic centers, and public healthcare systems, thereby contributing significantly to improved healthcare delivery and technological self-reliance.

In summary, the proposed SERS-based bacterial sepsis detection platform has the potential to improve clinical outcomes, reduce antimicrobial resistance, enhance diagnostic accessibility in resource-limited settings, and create strong commercial opportunities in the rapidly growing diagnostics market.

8. CHALLENGES & RISK FACTORS

Although the proposed SERS-based bacterial sepsis diagnostic platform shows strong potential, several technical, biological, and translational challenges must be addressed before clinical deployment.

A major technical challenge is achieving **reproducible and uniform SERS substrates**. Since signal enhancement depends on nanoscale plasmonic hotspots, even small variations in nanoparticle size, distribution, or surface morphology can cause large signal fluctuations. Maintaining batch-to-batch consistency during large-scale fabrication remains difficult and requires further optimization and strict quality control.

Another critical challenge is **interference from complex biological matrices** such as blood and plasma. These fluids contain a variety of proteins, lipids, and metabolites that can introduce background noise, peak overlap, and signal suppression. Non-specific adsorption of biomolecules onto the SERS surface can also reduce effective sensitivity. Improved sample preparation techniques and surface modifications are required to minimize these effects.

Sensitivity under real clinical conditions is another concern, as biomarker concentrations (PGN, LPS, LTA) may be very low during early stages of infection. Further enhancement of substrate performance and biomarker adsorption efficiency is necessary to reach clinically relevant detection limits.

Clinical sample **variability across patients and disease stages** also affects spectral consistency. Large-scale validation using diverse clinical samples will be required to ensure robustness and generalizability of the diagnostic system.

In addition, **spectral complexity and peak overlap** create challenges for data analysis, necessitating robust preprocessing and machine learning methods. However, building reliable models will require large, well-curated datasets.

Hardware-related challenges include **miniaturization and stability of portable Raman systems**, as compact devices may suffer from reduced signal quality and environmental noise. Careful optical and system-level design will be essential for point-of-care deployment.

Finally, **regulatory approval and cost scalability** are major translational hurdles. The system must meet stringent clinical regulations and remain affordable for widespread use, especially in public healthcare settings. Long-term stability of substrates, especially against silver oxidation, must also be addressed for real-world reliability.

In summary, while promising, the platform faces challenges related to substrate reproducibility, biological interference, sensitivity, clinical variability, data complexity, hardware integration, regulatory approval, and economic scalability, all of which must be systematically addressed for successful clinical translation.

9. ANY PATENT FILED

Currently, no patent has been filed. Patent filing is planned after experimental validation and device prototype development.

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